

DSpark audio draft head — vLLM serving results (Qwen3-Omni ASR)

Table 1. Single-request performance (concurrency 1, 12 utts)

Method	Accept. len τ (/8)	Throughput (tok/s)	TPOT (ms)	Latency (s, 12 utt.)
Autoregressive (no draft)	1.00	26.7	37.5	25.09
+ DSpark draft (block 7)	6.84	136.1	7.35	4.92
Speedup	6.8×	5.1×	5.1×	5.1×

Table 2. Throughput and latency under concurrency (32 reqs / level)

Concurrency	Baseline (tok/s)	DSpark (tok/s)	Speedup	Mean lat. (s)	p99 lat. (s)	τ (/8)
1	29.3	145.4	5.0×	0.39	0.58	7.00
4	95.4	494.6	5.2×	0.44	0.68	6.98
8	214.8	1057.7	4.9×	0.39	0.51	6.98
16	388.7	1203.0	3.1×	0.56	0.92	7.01

Qwen3-Omni-30B-A3B Thinker + DSpark audio draft (block 7) · LibriSpeech train-clean-100 · greedy · 1xB300 (TP=2) · vLLM 0.25.0.
Output verified identical token-for-token between configs. Requires the speculators config-loader fix (algos.py: dspark_bonus_anchor = not sample_from_anchor).